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Onderzoek de convergentie van de reeks $\sum_{n=0}^{\infty} \frac{n^2}{2^n}$

d'Alembert

$$\text{Stel } a_n = \frac{n^2}{2^n}$$

$$\text{Stel } a_{n+1} = \frac{(n+1)^2}{2^{n+1}}$$

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \\ &= \lim_{n \rightarrow \infty} \left(\frac{(n+1)^2}{2^{n+1}} \cdot \frac{2^n}{n^2} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{(n+1)^2}{n^2} \cdot \frac{2^n}{2 \cdot 2^n} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{n+1}{n} \right)^2 \cdot \frac{1}{2} \\ &= 1^2 \cdot \frac{1}{2} = \frac{1}{2} \end{aligned}$$

Aangezien $L = \frac{1}{2} < 1$, is de reeks convergent.

References